

Joseph Babu

📍 Kerala, India ✉ jb16042006@gmail.com
🌐 github.com/josephbabu1 🔗 linkedin.com/in/joseph-babu 🌐 josephbabu1.github.io

EDUCATION

Government Engineering College, Thrissur Aug 2024 – Present
B.Tech in Electronics and Communication Engineering
Bharatiya Vidya Bhavan, Poochatty 2022 – 2024
Higher Secondary Education
Relevant Coursework: Digital Electronics, Microprocessors, Analog Circuits, Data Structures, Python Programming, DSP, FPGA Design, Embedded Systems

TECHNICAL SKILLS

Languages: Python, C, Verilog, HTML/CSS/JavaScript, LaTeX
Tools: Git/GitHub, MATLAB, Simulink, Vivado, ModelSim, Keil μ Vision, Proteus, MCP2515, ESP32 /Arduino
Areas: Embedded Systems, FPGA Design, Digital Signal Processing, PCB Design, CAN Protocol, Hardware Security

EXPERIENCE

Formula Kart Racing Championship – Season 06 Jan 2026 – Mar2026
CAN Telemetry System Developer (EV Go-Kart) ESP32, CAN, C

- Developed complete CAN bus telemetry system for electric go-kart using ESP32 + MCP2515
- Read motor data (RPM, temperature, voltage) from VESC controller via CAN (250 kbps, extended frames)
- Displayed real-time metrics (speed from RPM, wheel diameter 27.94 cm) on 16×2 I2C LCD

TinkerHub, ISTE 2024 – Present

- Participating in technical workshops and hardware projects

PROJECTS

Chaos-Driven Time-Varying Convolutional Codec MATLAB, FPGA, Verilog

- Designed rate-1/2, K=7 convolutional encoder with dynamic generator polynomials using logistic map chaos
- Implemented 64-state Viterbi decoder with synchronized chaos for lightweight physical layer security
- Synthesizable Verilog for FPGA (Vivado/ModelSim); BER comparison with NASA (133,171) code

Digital Voltmeter (ADC0808 + 8051) Embedded C, Proteus, Keil

- 3-channel voltmeter using 8051 and ADC0808, 0-5V input, 19.5 mV resolution, error < 2%
- Programmed in Embedded C (Keil μ Vision), simulated in Proteus with 16×2 LCD (4-bit mode)

WiFi Controlled Car ESP32, C++, WiFi

- ESP32-based remote-controlled car with WiFi connectivity
- Complete IoT project demonstrating embedded controls and wireless communication

FPGA-Based Crypto Accelerator (AES-128 + SHA-256) Verilog, FPGA, Security

- Designing hardware accelerator for AES-128 encryption/decryption and SHA-256 hashing on FPGA

- ESP32 acts as USB-to-SPI bridge; Python script on PC sends encryption or HMAC commands
- Planned: AES-128 (ECB/CBC modes, pipelined), SHA-256 HMAC, write-only key storage, Xilinx Artix-7 target

OTHER PROJECTS

- **Pseudo-Random Sequence Generator:** Designed synchronous FPGA-based sequence generator with timed state transitions and conditional output logic
- **Digital Door Lock:** Implemented sequence-based authentication system with real-time validation and deterministic state transitions
- **Variable Voltage Regulator (LM723):** Designed 0–20V adjustable power supply using LM723 voltage regulator IC with current limiting and short-circuit protection.

ACHIEVEMENTS & HIGHLIGHTS

- **Research in Hardware Security:** Active work on chaos-based coding and AES-128 hardware accelerator
- **Team Recognition:** CAN telemetry system used in official Formula Kart Racing Championship
- **Technical Breadth:** Proficient across FPGA (Verilog), embedded microcontrollers (8051, ESP32), and software (Python, web)
- **Certifications:** Flutter Development, Electric Vehicle Orientation, Digital Marketing

INTERESTS & ACTIVITIES

- Embedded systems, FPGA design, DSP, hardware security
- Open-source contributions and automation tools
- Circuit design and simulation